

Sharp Simultaneous Shadows and Fiber Realization for Hölder–RL Relations

Author information to be supplied

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Abstract

Let H be a real Hilbert space and let $F : H \rightrightarrows H$ satisfy $\|(x - y) - \lambda(v - w)\| \leq L\|(x - y) + \lambda(v - w)\|^\gamma$ for all graph pairs, where $\lambda, L > 0$ and $0 < \gamma < 1$. We prove that one strongly monotone bi-Lipschitz homeomorphism $A : H \rightarrow H$ simultaneously approximates every nonempty forward fiber $F(x)$ and inverse fiber $F^{-1}(v)$. Writing $\mathcal{R} = L^{1/(1-\gamma)}$, the respective uniform errors are $\mathcal{R}/(\sqrt{2}\lambda)$ and $\mathcal{R}/\sqrt{2}$, and the dimension-independent factor $1/\sqrt{2}$ is optimal. The construction also provides a cross inequality and an explicit monotone-enlargement inclusion. Preserving a specified graph point changes the optimal factor to 1. In finite dimensions, for relations that are graph-maximal at the fixed parameters, the possible complete inverse fibers are exactly the nonempty compact sets of diameter at most $\mathcal{R}$, with a scaled statement for forward fibers. The realization attains the smallest Hölder constant permitted by the prescribed set. We establish sharp range localization in fixed windows and give a finite-data quadratic-program construction of globally invertible surrogates with coverage-dependent paired error certificates. A deadband model illustrates stable approximate inversion without claiming stable exact recovery. The results distinguish global simultaneous organization from the unrestricted compact geometry possible within an individual fiber.

Keywords. Set-valued relation; Hölder continuity; strong monotonicity; Kirschbraun extension; fiber realization; certified approximation.

Contents

1	Introduction	2
2	Relations, graph coordinates, and fixed parameters	4
3	Sharp simultaneous strongly monotone shadows	5
3.1	Quadratic excess and orthogonal lifting	5
3.2	The main approximation theorem	6
3.3	Dimension-independent optimality	7
3.4	Preserving a specified graph point	8
4	Finite-dimensional geometry and exact fiber realization	9
4.1	Full domain, full range, and compact fibers	9
4.2	Prescribed fixed-point sets with optimal Hölder modulus	10
4.3	A single-valued Lipschitz realization	12
5	Sharp range localization in global and fixed windows	13

6	Finite-data shadows and coverage certificates	14
6.1	A globally consistent quadratic-program construction	14
6.2	Coverage-dependent paired certificates	16
6.3	Finite-precision evaluation	17
6.4	Why a finite total transcript cannot certify the whole space	18
7	A deadband inverse problem: what the certificate changes	18
8	Discussion and scope	20

1 Introduction

A relation may assign several outputs to one input and several inputs to one output. Describing an individual value set does not necessarily explain how different value sets fit together. Conversely, the existence of a well-conditioned approximation need not restrict the topology of the original fibers. This paper studies these complementary questions for relations satisfying a sublinear Hölder condition in graph coordinates.

Throughout, H is a real Hilbert space and $\lambda, L > 0$ and $0 < \gamma < 1$ are fixed. We consider the all-pairs condition

$$\|(x - y) - \lambda(v - w)\| \leq L\|(x - y) + \lambda(v - w)\|^\gamma \quad ((x, v), (y, w) \in \text{gph } F). \quad (1.1)$$

We call this the Hölder–RL condition, where the terminology is only a label for (1.1). It is a global condition on all graph pairs, not a condition only relative to a solution or along an individual branch. The natural length scale is $\mathcal{R} = L^{1/(1-\gamma)}$.

Our first result concerns approximation at the *same original base point*. Ordinary graph proximity allows both coordinates of a graph point to move. In contrast, a forward fiber estimate holds with the input fixed, and an inverse fiber estimate holds with the output fixed. We construct one strongly monotone bi-Lipschitz homeomorphism $A : H \rightarrow H$ satisfying

$$\sup_{v \in F(x)} \|v - A(x)\| \leq \frac{\mathcal{R}}{\sqrt{2}\lambda}, \quad \sup_{x \in F^{-1}(v)} \|x - A^{-1}(v)\| \leq \frac{\mathcal{R}}{\sqrt{2}}, \quad (1.2)$$

whenever the indicated fiber is nonempty. The same map works for every input and every output. The factor $1/\sqrt{2}$ is optimal uniformly over dimensions, even if the approximating center is not required to come from a monotone map. We call such an A a *shadow*. This word does not mean that A is a selection of F : its graph need not lie in $\text{gph } F$.

The proof passes through the partial map $C(x + \lambda v) = x - \lambda v$. A Hölder estimate bounds its excess above a chosen Lipschitz slope. Orthogonal lifting and the classical Kirszbraun theorem then give a contraction N with the two-parameter estimate

$$\|C(p) - N(q)\|^2 \leq \sigma^2 \|p - q\|^2 + M_\sigma/2, \quad p \in \text{dom } C, \ q \in H.$$

It is this cross estimate, rather than just a same-parameter uniform approximation bound, that yields (1.2) without a condition-number loss. Global graph maximality is not needed for this part of the theory.

Our second main result determines the possible complete fibers of graph-maximal relations in finite dimensions. At fixed parameters, a set $K \subset \mathbb{R}^n$ occurs as $F^{-1}(0)$ if and only if it is nonempty, compact, and has diameter at most $\mathcal{R}$. The forward-fiber threshold is $\mathcal{R}/\lambda$. The sufficient direction

constructs a map with prescribed fixed-point set K and Hölder constant $(\text{diam } K)^{1-\gamma}$, the least constant compatible with that set. The construction can also make the original F single-valued and globally Lipschitz. Thus a common stable shadow is compatible with arbitrary compact fiber geometry at the permitted scale. The realization theorem is a positive classification, not merely a collection of examples showing failures of convexity.

We give two further quantitative consequences. First, an exact scalar localization function gives sharp ball-to-ball range coverage, including the setting where graph maximality is known only inside a fixed window. Second, finitely many graph observations produce one globally consistent contraction through a strongly convex quadratic program. Parameter-space coverage turns sampled estimates into paired certificates for unobserved graph points. A finite-information obstruction explains why finitely many observations cannot certify a uniform error on an unrestricted unbounded domain. A deadband inverse problem illustrates the precise tradeoff: stable approximate representatives can coexist with unstable exact inverse selections. The certificate does not recover information absent from the original model.

Relation to earlier work. Luke and Tam [7, Theorem 1] study generalized monotonicity and the proximal point algorithm, including representations of value sets and inverse value sets by lower-level sets of lower semicontinuous weakly convex functions in their relatively maximal submonotone framework. Their original- relation viewpoint motivates our questions. We consider a different, sublinear all-pairs condition and ask for simultaneous quantitative approximation and exact fiber realization. We do not claim to settle the Bregman, stochastic, or algorithmic directions in their outlook.

The tools underlying our proofs are classical. The Kirszbraum–Valentine theorem [5, 11] supplies Hilbert-space Lipschitz extension. Same-constant Hölder extension follows from the Hilbert snowflake embedding associated with Schoenberg’s theorem [10], and is discussed together with monotone extension by Minty [8]. Uniform Lipschitz approximation of uniformly continuous Hilbert-valued mappings is already established by Levy and Rice [6]. Constructive links between Lipschitz extension and monotone operator theory are developed, for example, by Bauschke and Wang [2]. We do not claim these approximation or extension principles as new. The conclusion here concerns the original relation and its inverse at fixed original base points, one common strongly monotone homeomorphism, and a dimension-independent optimal constant. The enlargement consequence uses the established notion of monotone enlargement [4].

The finite-data formula is a specialization of constructive Kirszbraum extension. Azagra, Le Gruyer, and Mudarra [1, Theorems 2 and 8] give an explicit Lipschitz extension formula and an extension theorem for strongly bi-Lipschitz maps. Algorithmic Kirszbraum extension for vector-valued regression is studied by Zaichyk et al. [12]. We provide a direct proof of the particular quadratic program used here, rather than claiming a new general extension algorithm.

Exact interpolation within prescribed operator classes is a related but different question. Rubbens, Hendrickx, and Taylor [9, Section 3.3] develop strengthened finite-data conditions for strongly monotone Lipschitz operators, including necessary relaxations of one-point extendibility. Here the original RL samples need not be exactly interpolable by that target class. Our construction accepts a quantified approximation error and provides paired, coverage-dependent certificates for the same surrogate of the underlying relation. Classical extension, contraction, and degree arguments are identified as tools rather than additional novelty claims.

The present paper is a structure and approximation study. It does not assert that a shadow together with a residual error bound automatically implies convergence of an algorithm for the original relation.

2 Relations, graph coordinates, and fixed parameters

For $F : H \rightrightarrows H$, write

$$\text{gph } F = \{(x, v) : v \in F(x)\}, \quad \text{dom } F = \{x : F(x) \neq \emptyset\}, \quad \text{ran } F = \{v : F^{-1}(v) \neq \emptyset\},$$

where $F^{-1}(v) = \{x : v \in F(x)\}$. A *fiber* is one of these forward or inverse values. An open ball is denoted by $B(x, r)$ and a closed ball by $\overline{B}(x, r)$. For nonempty sets E, D , their Hausdorff distance is

$$d_H(E, D) = \max\left\{\sup_{e \in E} \text{dist}(e, D), \sup_{d \in D} \text{dist}(d, E)\right\},$$

with the value $+\infty$ allowed. In particular, $d_H(E, \{a\}) = \sup_{e \in E} \|e - a\|$.

Definition 2.1. A nonempty relation F is a (λ, L, γ) Hölder–RL relation if (1.1) holds. It is *graph-maximal at the fixed parameters* if no proper superset of its graph satisfies that same inequality with the same λ, L, γ . For $U, W \subset H$, a nonempty graph $G \subset U \times W$ is *relatively maximal* if no proper superset within $U \times W$ satisfies the fixed-parameter inequality.

This maximality is not maximal monotonicity. A map $A : H \rightarrow H$ is μ -strongly monotone if $\langle A(x) - A(y), x - y \rangle \geq \mu \|x - y\|^2$ for all x, y . It is ℓ -Lipschitz if $\|A(x) - A(y)\| \leq \ell \|x - y\|$. A bi-Lipschitz homeomorphism here means a bijection whose map and inverse are Lipschitz.

Lemma 2.2 (Graph coordinates). *Let F satisfy (1.1) and put*

$$D = \{x + \lambda v : (x, v) \in \text{gph } F\}, \quad C(x + \lambda v) = x - \lambda v. \quad (2.1)$$

Then $C : D \rightarrow H$ is well-defined and L -Hölder with exponent γ . Conversely, any such map defines an RL graph by

$$\text{gph } F = \left\{ \left(\frac{p + C(p)}{2}, \frac{p - C(p)}{2\lambda} \right) : p \in D \right\}. \quad (2.2)$$

If $D = H$, this graph is graph-maximal at the fixed parameters.

Proof. If two graph points have the same sum $x + \lambda v$, the right-hand side of (1.1) is zero. Their differences $x - \lambda v$ are therefore equal. Adding and subtracting the equalities identifies the two original points. The RL inequality is exactly $\|C(p) - C(q)\| \leq L \|p - q\|^\gamma$, and solving for (x, v) gives (2.2). Conversely, substitution into (1.1) verifies the condition. If $D = H$ and a pair (x, v) is compatible with the entire graph, compare it with the existing pair at parameter $p = x + \lambda v$. Their parameter difference is zero, so compatibility forces their second graph coordinate to agree. The proposed pair is already in the graph, proving maximality. $\square$

We use two classical extension results. The Kirszbraun–Valentine theorem states that a Lipschitz map from a subset of a Hilbert space into another Hilbert space has a same-constant extension to the whole domain space [5, 11]. We also use the following Hölder consequence.

Lemma 2.3 (Classical Hölder extension). *Every L -Hölder map $C : D \subset H \rightarrow H$ of exponent $0 < \gamma < 1$ has an L -Hölder extension $\widehat{C} : H \rightarrow H$ of the same exponent. Consequently an RL relation is graph-maximal if and only if $D = H$ in (2.1). Every RL graph has a graph-maximal completion at the same parameters.*

Proof. The Hilbert snowflake theorem [10] gives a Hilbert space E and a map $J : H \rightarrow E$ such that $\|J(p) - J(q)\|_E = \|p - q\|^\gamma$. The map $J(p) \mapsto C(p)$ on $J(D)$ is L -Lipschitz, so Kirszbraun extension to E , followed by composition with J , gives $\widehat{C}$. This proves the extension statement (see also [8]). If an RL graph were maximal with $D \neq H$, (2.2) applied to $\widehat{C}$ would give a proper compatible extension, a contradiction. The converse is Lemma 2.2, and the same construction proves the completion assertion. $\square$

Lemma 2.4 (Fiber diameters). *Every RL relation satisfies*

$$\text{diam } F(x) \leq \mathcal{R}/\lambda, \quad \text{diam } F^{-1}(v) \leq \mathcal{R} \quad (2.3)$$

for each nonempty fiber. Neither maximality nor closedness is needed.

Proof. For $v, w \in F(x)$, set $d = \lambda\|v - w\|$. Then $d \leq Ld^\gamma$. If $d > 0$, division gives $d \leq L^{1/(1-\gamma)} = \mathcal{R}$; the case $d = 0$ is immediate. Setting the outputs equal proves the inverse assertion. $\square$

3 Sharp simultaneous strongly monotone shadows

3.1 Quadratic excess and orthogonal lifting

For $\sigma > 0$, define

$$M_\sigma = (1 - \gamma)\gamma^{\gamma/(1-\gamma)}\mathcal{R}^2\sigma^{-2\gamma/(1-\gamma)}. \quad (3.1)$$

Lemma 3.1 (Exact quadratic excess). *For every $\sigma > 0$,*

$$\sup_{t \geq 0} \{L^2 t^{2\gamma} - \sigma^2 t^2\} = M_\sigma. \quad (3.2)$$

The maximum is attained at $t_\sigma = (\gamma L^2 / \sigma^2)^{1/[2(1-\gamma)]}$.

Proof. For $t > 0$ the derivative vanishes precisely when $\gamma L^2 t^{2\gamma-2} = \sigma^2$. The function is positive for sufficiently small positive t , equals zero at zero, and tends to $-\infty$ at infinity. The unique positive critical point is therefore the maximum. Since $\sigma^2 t_\sigma^2 = \gamma L^2 t_\sigma^{2\gamma}$, substitution gives $(1 - \gamma)L^2 t_\sigma^{2\gamma}$, which simplifies to (3.1). $\square$

Lemma 3.2 (Cross-lift). *Let $C : D \subset H \rightarrow H$ be L -Hölder of exponent γ and $D \neq \emptyset$. For every $0 < \sigma < 1$ there exists a σ -Lipschitz map $N : H \rightarrow H$ such that*

$$\|C(p) - N(q)\|^2 \leq \sigma^2 \|p - q\|^2 + M_\sigma/2 \quad (p \in D, q \in H). \quad (3.3)$$

Proof. Let $\ell_2(D)$ be the Hilbert space of square-summable families indexed by the set D , with coordinate unit vectors e_p , and set $r^2 = M_\sigma/(2\sigma^2)$. On $\widehat{D} = \{(p, re_p) : p \in D\} \subset H \oplus \ell_2(D)$ define $\widehat{C}(p, re_p) = C(p)$. For distinct $p, s \in D$,

$$\|C(p) - C(s)\|^2 \leq L^2 \|p - s\|^{2\gamma} \leq \sigma^2 \|p - s\|^2 + M_\sigma = \sigma^2 \|(p, re_p) - (s, re_s)\|^2.$$

Thus $\widehat{C}$ is σ -Lipschitz; the identical-point case is automatic. Kirszbraun's theorem gives a σ -Lipschitz extension $\widehat{N} : H \oplus \ell_2(D) \rightarrow H$. Set $N(q) = \widehat{N}(q, 0)$. Comparing a lifted point with a base-plane point gives

$$\|C(p) - N(q)\|^2 \leq \sigma^2 (\|p - q\|^2 + r^2) = \sigma^2 \|p - q\|^2 + M_\sigma/2,$$

as claimed. Possible nonseparability of $\ell_2(D)$ does not change the Hilbert-space extension theorem. $\square$

3.2 The main approximation theorem

Theorem 3.3 (One shadow in both directions). *Let F be a nonempty (λ, L, γ) RL relation on a real Hilbert space. For each $0 < \sigma < 1$ there exists a bi-Lipschitz homeomorphism $A_\sigma : H \rightarrow H$ satisfying*

$$\text{Lip}(A_\sigma) \leq \frac{1 + \sigma}{\lambda(1 - \sigma)}, \quad \text{Lip}(A_\sigma^{-1}) \leq \frac{\lambda(1 + \sigma)}{1 - \sigma}, \quad (3.4)$$

$$\langle A_\sigma(x) - A_\sigma(y), x - y \rangle \geq \frac{1 - \sigma}{\lambda(1 + \sigma)} \|x - y\|^2. \quad (3.5)$$

For every $(x, v) \in \text{gph } F$ and $z \in H$, the same map satisfies

$$\langle x - z, v - A_\sigma(z) \rangle \geq \frac{1 - \sigma^2}{4\lambda} \|x - z + \lambda(v - A_\sigma(z))\|^2 - \frac{M_\sigma}{8\lambda}. \quad (3.6)$$

In particular, with $r_\sigma = \sqrt{M_\sigma/[2(1 - \sigma^2)]}$,

$$\|v - A_\sigma(x)\| \leq r_\sigma/\lambda, \quad \|x - A_\sigma^{-1}(v)\| \leq r_\sigma \quad ((x, v) \in \text{gph } F). \quad (3.7)$$

The displayed radius r_σ is minimized at $\sigma = \sqrt{\gamma}$. For the resulting map A ,

$$\|v - A(x)\| \leq \frac{\mathcal{R}}{\sqrt{2}\lambda}, \quad \|x - A^{-1}(v)\| \leq \frac{\mathcal{R}}{\sqrt{2}} \quad ((x, v) \in \text{gph } F). \quad (3.8)$$

Proof. Apply Lemma 3.2 directly to the partial map in Lemma 2.2, and define

$$X(q) = \frac{q + N(q)}{2}, \quad V(q) = \frac{q - N(q)}{2\lambda}. \quad (3.9)$$

For any $x \in H$, the equation $X(q) = x$ is equivalent to $q = 2x - N(q)$. Its right-hand side is a contraction of modulus at most σ , so the Banach fixed-point theorem gives a unique solution. For any $v \in H$, the same theorem applied to $q = 2\lambda v + N(q)$ gives a unique solution of $V(q) = v$. Thus both X and V are bijections and $A_\sigma = V \circ X^{-1}$ is a bijection.

For $q_1, q_2 \in H$, put $h = q_1 - q_2$ and $k = N(q_1) - N(q_2)$, so that $\|k\| \leq \sigma\|h\|$. Differences in the original coordinates are $(h + k)/2$ and $(h - k)/(2\lambda)$. The estimates

$$(1 - \sigma)\|h\| \leq \|h \pm k\| \leq (1 + \sigma)\|h\|$$

give both bounds in (3.4). Moreover,

$$\begin{aligned} \langle V(q_1) - V(q_2), X(q_1) - X(q_2) \rangle &= \frac{\|h\|^2 - \|k\|^2}{4\lambda} \\ &\geq \frac{1 - \sigma^2}{4\lambda} \|h\|^2 \geq \frac{1 - \sigma}{\lambda(1 + \sigma)} \|X(q_1) - X(q_2)\|^2. \end{aligned}$$

This proves (3.5); the Lipschitz bounds also establish the asserted homeomorphism property.

For an original graph point and a point on the shadow, let $p = x + \lambda v \in D$ and $q = z + \lambda A_\sigma(z)$. Then

$$p - q = (x - z) + \lambda(v - A_\sigma(z)), \quad C(p) - N(q) = (x - z) - \lambda(v - A_\sigma(z)).$$

The difference of their squared norms is $4\lambda\langle x - z, v - A_\sigma(z) \rangle$. Therefore (3.3) gives (3.6). Taking $z = x$ makes its left side zero and gives the forward bound in (3.7). Taking $z = A_\sigma^{-1}(v)$ gives the inverse bound. These two substitutions use the same A_σ .

Finally, putting $u = \sigma^2$, the squared radius is a positive σ -independent constant times $u^{-\gamma/(1-\gamma)}(1-u)^{-1}$. Its logarithmic derivative is

$$-\frac{\gamma}{(1-\gamma)u} + \frac{1}{1-u},$$

which vanishes only at $u = \gamma$. The function diverges at both endpoints, so this point is the unique minimum. Substitution gives $M_{\sqrt{\gamma}} = (1-\gamma)\mathcal{R}^2$ and $r_{\sqrt{\gamma}}^2 = \mathcal{R}^2/2$, proving (3.8). $\square$

The quantifiers in this theorem matter: one A controls all graph points. In an arbitrary Hilbert space the theorem says nothing about the nonemptiness of a fiber not already represented in $\text{gph } F$.

Corollary 3.4 (Enlargement inclusion). *The map A_σ in Theorem 3.3 is maximal monotone. With the standard enlargement*

$$A_\sigma^{[\varepsilon]}(x) = \{v : \langle x - z, v - A_\sigma(z) \rangle \geq -\varepsilon \text{ for every } z \in H\},$$

one has

$$F(x) \subset A_\sigma^{[M_\sigma/(8\lambda)]}(x). \quad (3.10)$$

For the map minimizing the radius in (3.7), $\varepsilon = (1-\gamma)\mathcal{R}^2/(8\lambda)$.

Proof. Strong monotonicity implies monotonicity. If (u, w) is monotonically related to the entire graph of the continuous full-domain map A_σ , then, for $z = u + th$ and $t > 0$, $\langle h, w - A_\sigma(u + th) \rangle \leq 0$. Letting $t \downarrow 0$ and then replacing h by $-h$ gives $\langle h, w - A_\sigma(u) \rangle = 0$ for every h , whence $w = A_\sigma(u)$. Thus no point can be added monotonically. Dropping the nonnegative squared term in (3.6) proves the inclusion, and substituting $\sigma = \sqrt{\gamma}$ gives the stated parameter. $\square$

The displayed enlargement parameter is associated with the map that minimizes the fiber-radius guarantee. We do not assert that it is the smallest possible enlargement parameter.

3.3 Dimension-independent optimality

Theorem 3.5 (Optimal universal factor). *The factor $1/\sqrt{2}$ in each inequality of (3.8) cannot be replaced by a smaller constant valid in all finite dimensions. This remains true for graph-maximal RL relations and arbitrary choices of a single center for the fiber under consideration.*

Proof. Fix $n \geq 1$ and take vertices $u_0, \dots, u_n \in \mathbb{R}^n$ of a regular simplex of edge length $\mathcal{R}$, centered so that $\sum_i u_i = 0$. Let $Q = \text{conv}\{u_0, \dots, u_n\}$. To compute the vertex radius, use

$$\sum_{j=0}^n \|u_i - u_j\|^2 = (n+1)\|u_i\|^2 + \sum_{j=0}^n \|u_j\|^2.$$

The left side is $n\mathcal{R}^2$ for every i , so all vertex norms are equal and their common squared value is $r_n^2 = n\mathcal{R}^2/[2(n+1)]$. For any $b \in \mathbb{R}^n$,

$$\frac{1}{n+1} \sum_{i=0}^n \|u_i - b\|^2 = r_n^2 + \|b\|^2.$$

At least one vertex is therefore at distance at least r_n from b .

The metric projection P_Q is nonexpansive and has image of diameter $\mathcal{R}$. Consequently

$$\|P_Q(p) - P_Q(q)\| \leq \min\{\|p - q\|, \mathcal{R}\} \leq \mathcal{R}^{1-\gamma} \|p - q\|^\gamma = L \|p - q\|^\gamma.$$

For the middle inequality, if $t \leq \mathcal{R}$ then $t \leq \mathcal{R}^{1-\gamma}t^\gamma$, while if $t \geq \mathcal{R}$ then $\mathcal{R} \leq \mathcal{R}^{1-\gamma}t^\gamma$.

Use $C = -P_Q$ in (2.2) with domain $\mathbb{R}^n$. The resulting relation is graph-maximal by Lemma 2.2. Its input is zero exactly when $p = P_Q p$, or $p \in Q$, and then its output is p/λ . Thus $F(0) = Q/\lambda$. Any center b has worst-case distance at least r_n/λ from this fiber. Any universal forward factor c must therefore satisfy $c \geq \sqrt{n/[2(n+1)]}$ for every n , so $c \geq 1/\sqrt{2}$.

Using $C = P_Q$ instead yields $F^{-1}(0) = Q$ and gives the same lower bound for any inverse-fiber center. The two witnesses need not be the same relation: either one already rules out a smaller common universal factor. $\square$

This theorem does not determine the best factor in each fixed dimension, nor does it assert optimality of the conditioning bounds in (3.4)–(3.5).

3.4 Preserving a specified graph point

Theorem 3.6 (Anchored shadow). *Under the assumptions of Theorem 3.3, fix $(x_0, v_0) \in \text{gph } F$. For every $0 < \sigma < 1$ there exists a bi-Lipschitz homeomorphism $A_{\sigma,0} : H \rightarrow H$ with the conditioning and strong-monotonicity bounds (3.4)–(3.5), such that $A_{\sigma,0}(x_0) = v_0$ and, for every $(x, v) \in \text{gph } F$ and $z \in H$,*

$$\langle x - z, v - A_{\sigma,0}(z) \rangle \geq \frac{1 - \sigma^2}{4\lambda} \|x - z + \lambda(v - A_{\sigma,0}(z))\|^2 - \frac{M_\sigma}{4\lambda}. \quad (3.11)$$

Writing $A_0 = A_{\sqrt{\gamma},0}$, this gives, for every original graph point,

$$\|v - A_0(x)\| \leq \mathcal{R}/\lambda, \quad \|x - A_0^{-1}(v)\| \leq \mathcal{R}, \quad (3.12)$$

and $F(x) \subset A_0^{[(1-\gamma)\mathcal{R}^2/(4\lambda)]}(x)$. The factor 1 is optimal under the anchoring requirement, already in dimension one.

Proof. Let $p_0 = x_0 + \lambda v_0$ and $c_0 = x_0 - \lambda v_0$. Repeat the lift of Lemma 3.2 with $r^2 = M_\sigma/\sigma^2$, and add the base datum $(p_0, 0) \mapsto c_0$. Distinct lifted data are compatible with Lipschitz modulus σ because their extra squared distance supplies $2M_\sigma$, at least the excess required by Lemma 3.1. A lifted datum and the anchor are compatible because

$$\|C(p) - c_0\|^2 \leq \sigma^2 \|p - p_0\|^2 + M_\sigma = \sigma^2 \|(p, re_p) - (p_0, 0)\|^2.$$

Kirszbraun extension now gives a contraction N with $N(p_0) = c_0$ and

$$\|C(p) - N(q)\|^2 \leq \sigma^2 \|p - q\|^2 + M_\sigma.$$

Construct $A_{\sigma,0}$ as in (3.9). The anchor equalities give $X(p_0) = x_0$ and $V(p_0) = v_0$. The conditioning proof is unchanged, and the difference-of-squares identity gives (3.11). Setting $z = x$ or $z = A_{\sigma,0}^{-1}(v)$ yields forward and inverse radii $\lambda^{-1}\sqrt{M_\sigma/(1-\sigma^2)}$ and $\sqrt{M_\sigma/(1-\sigma^2)}$, respectively. At $\sigma = \sqrt{\gamma}$ these equal $\mathcal{R}/\lambda$ and $\mathcal{R}$. Dropping the square gives the stated enlargement inclusion.

For forward optimality in $\mathbb{R}$, take $Q = [0, \mathcal{R}]$, use $C = -P_Q$, and anchor at $(0, 0)$. Then $F(0) = [0, \mathcal{R}/\lambda]$, whereas the anchoring requires $A_0(0) = 0$. Its worst-case forward error is at least $\mathcal{R}/\lambda$. For inverse optimality use $C = P_Q$, so $F^{-1}(0) = Q$, and the same anchor forces $A_0^{-1}(0) = 0$. The worst-case inverse error is at least $\mathcal{R}$. $\square$

4 Finite-dimensional geometry and exact fiber realization

4.1 Full domain, full range, and compact fibers

Theorem 4.1. *Let $H = \mathbb{R}^n$ and let F be graph-maximal at the fixed RL parameters. Then*

$$\text{dom } F = \text{ran } F = \mathbb{R}^n. \quad (4.1)$$

Every forward and inverse fiber is compact. Both coordinate projections from $\text{gph } F$ are proper, and F and F^{-1} are upper semicontinuous and locally bounded. They map compact sets to compact unions of values. Orient the graph by the parametrization of Lemma 2.2, with the standard orientation on $\mathbb{R}^n$. Under this convention, each projection has degree 1.

Proof. By Lemma 2.3, the Cayley map C is defined on all of $\mathbb{R}^n$. It is continuous and satisfies $\|C(p)\| \leq \|C(0)\| + L\|p\|^\gamma$. The map $p \mapsto ((p + C(p))/2, (p - C(p))/(2\lambda))$ is a homeomorphism onto the graph, with continuous inverse $(x, v) \mapsto x + \lambda v$. For $t \in [0, 1]$ define

$$X_t(p) = \frac{p + tC(p)}{2}, \quad V_t(p) = \frac{p - tC(p)}{2\lambda}.$$

Uniformly in t ,

$$2\|X_t(p)\| \geq \|p\| - \|C(0)\| - L\|p\|^\gamma, \quad 2\lambda\|V_t(p)\| \geq \|p\| - \|C(0)\| - L\|p\|^\gamma.$$

The right sides tend to infinity with $\|p\|$. Hence both homotopies are uniformly proper: preimages of any bounded target set remain in a common bounded parameter set.

Fix a target y . Choose a large ball so that neither homotopy hits y on its boundary. Homotopy invariance of Brouwer degree gives degree 1 for X_1 and V_1 , since $X_0 = \text{Id}/2$ and $V_0 = \text{Id}/(2\lambda)$ are positive scalar multiples of the identity. The nonzero-degree existence property gives a preimage of y for each map, proving (4.1). These are classical degree properties; see, for example, [3].

The displayed coercive bounds and continuity show that the preimage of a compact set under either projection, expressed in the parameter p , is closed and bounded, hence compact. The projections are thus proper, and point preimages give compact fibers. The graph is closed: if $(x_j, v_j) \rightarrow (x, v)$, then $p_j = x_j + \lambda v_j \rightarrow x + \lambda v$ and $C(p_j) = x_j - \lambda v_j \rightarrow x - \lambda v$, which verifies graph membership by continuity. The same coercive bounds show local boundedness of each fiber map: bounded x bounds p , hence v , and bounded v bounds p , hence x .

For upper semicontinuity, suppose an open set O contains $F(x)$ but there are $x_j \rightarrow x$ and $v_j \in F(x_j) \setminus O$. Local boundedness gives a convergent subsequence of v_j ; closedness of the graph puts its limit in $F(x) \subset O$, contradicting openness. The inverse argument is identical with the coordinates exchanged. Finally, for a compact set K , the portion of the graph above K is compact by properness; its projection onto the other coordinate is $F(K)$ and is compact. The inverse statement follows from the other projection. $\square$

Corollary 4.2. *For a relation satisfying Theorem 4.1, the same shadow A satisfies the complete-fiber Hausdorff estimates*

$$\sup_{x \in \mathbb{R}^n} d_H(F(x), \{A(x)\}) \leq \frac{\mathcal{R}}{\sqrt{2}\lambda}, \quad \sup_{v \in \mathbb{R}^n} d_H(F^{-1}(v), \{A^{-1}(v)\}) \leq \frac{\mathcal{R}}{\sqrt{2}}.$$

Proof. All fibers are nonempty by (4.1), and Hausdorff distance to a singleton equals the supremum of the distances to that singleton. Apply (3.8). $\square$

Finite dimensionality in Theorem 4.1 is essential to this proof: bounded closed subsets of an infinite-dimensional Hilbert space need not be compact, and Brouwer degree is not available in that generality. We do not assert (4.1) for arbitrary Hilbert spaces.

4.2 Prescribed fixed-point sets with optimal Hölder modulus

Lemma 4.3 (A strict convex-hull margin). *Let K be a nonempty compact subset of a Hilbert space, let $D_K = \text{diam } K$, and let $Q = \overline{\text{conv}} K$. Then, for every $p \in Q$,*

$$\sup_{q \in Q} \|p - q\|^2 \leq D_K^2 - \text{dist}(p, K)^2. \quad (4.2)$$

Proof. Every $k \in K$ and $q \in Q$ satisfy $\|k - q\| \leq D_K$: this follows by convexity of the norm for a finite convex combination q , and then by passage to its limit. Choose finite convex combinations $p_m = \sum_i \alpha_i k_i \rightarrow p$, with $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$. For any fixed $q \in Q$, the Hilbert variance identity gives

$$\|p_m - q\|^2 = \sum_i \alpha_i \|k_i - q\|^2 - \sum_i \alpha_i \|k_i - p_m\|^2 \leq D_K^2 - \text{dist}(p_m, K)^2.$$

The distance function is continuous, so taking $m \rightarrow \infty$ proves the inequality for each q . Taking its supremum gives (4.2). $\square$

Theorem 4.4 (Optimal prescribed fixed-point realization). *Let K be a nonempty compact subset of a real Hilbert space and $D_K = \text{diam } K$. For every $0 < \gamma < 1$ there exists a map $R : H \rightarrow \overline{\text{conv}} K$ such that*

$$\text{Fix } R = K, \quad \|R(x) - R(y)\| \leq D_K^{1-\gamma} \|x - y\|^\gamma \quad (x, y \in H). \quad (4.3)$$

For $D_K = 0$, the coefficient is interpreted as zero. For $D_K > 0$, no smaller Hölder constant is possible for any map fixing every point of K .

Proof. If $D_K = 0$, the constant map to the sole point of K has all asserted properties. Assume $D_K > 0$, set $Q = \overline{\text{conv}} K$ and $L_0 = D_K^{1-\gamma}$, and fix $k_0 \in K$. If $Q = K$, use $R = P_Q$. Nonexpansiveness and the range diameter give

$$\|P_Q x - P_Q y\| \leq \min\{\|x - y\|, D_K\} \leq L_0 \|x - y\|^\gamma,$$

and the fixed-point set is $Q = K$.

Suppose $Q \setminus K \neq \emptyset$. For each $p \in Q \setminus K$, put $d_p = \text{dist}(p, K) > 0$ and $E_p = \sqrt{D_K^2 - d_p^2} < D_K$. By Lemma 4.3, $\|p - q\| \leq E_p$ for every $q \in Q$. Choose

$$0 < r_p < \min\{d_p/2, (D_K - E_p)/2\}, \quad D_p = E_p + r_p < D_K,$$

and define on Q

$$\phi_p(y) = \max\{1 - \|y - p\|/r_p, 0\}, \quad H_p(y) = \phi_p(y)(y - k_0), \quad M_p = 1 + D_K/r_p.$$

The function ϕ_p vanishes on K , is positive in the relative open ball $Q \cap B(p, r_p)$, and has Lipschitz constant at most $1/r_p$. For $y, z \in Q$,

$$\|H_p(y) - H_p(z)\| \leq \phi_p(y)\|y - z\| + |\phi_p(y) - \phi_p(z)|\|z - k_0\| \leq M_p\|y - z\|.$$

Choose a positive number

$$0 < \varepsilon_p \leq \min \left\{ 1, \frac{L_0 D_p^{\gamma-1} - 1}{M_p} \right\}. \quad (4.4)$$

The right side is positive because $D_p < D_K$ and $\gamma - 1 < 0$. Set $S_p(y) = y - \varepsilon_p H_p(y)$. It is a convex combination of y and k_0 , so $S_p(Q) \subset Q$, and it fixes K .

We check its Hölder modulus for all point pairs. If $\phi_p(y) = \phi_p(z) = 0$, then $S_p(y) - S_p(z) = y - z$ and $\|y - z\| \leq D_K$ gives the required L_0 -Hölder bound. Otherwise assume, after interchanging y, z if necessary, that $\phi_p(y) > 0$. Then $h = \|y - z\| < r_p + E_p = D_p$. If $h = 0$ there is nothing to prove; if $h > 0$, (4.4) gives

$$\|S_p(y) - S_p(z)\| \leq (1 + \varepsilon_p M_p)h \leq L_0 D_p^{\gamma-1} h \leq L_0 h^\gamma.$$

The last inequality uses the negative exponent $\gamma - 1$.

The compact metric space K is separable, and rational convex combinations of a countable dense subset make Q separable. Hence $Q \setminus K$ is second countable. Its cover by the relative open balls $Q \cap B(p, r_p)$ has a finite or countable subcover. Enumerate the associated maps as S_j , and choose strictly positive weights a_j summing to one. Define

$$S(y) = \sum_j a_j S_j(y).$$

The series converges uniformly because Q is bounded. Its sum lies in Q : finite weighted sums completed with the remaining weight at k_0 lie in Q and converge to the same limit. The common Hölder estimate passes to the sum. Moreover,

$$S(y) = y - \theta(y)(y - k_0), \quad \theta(y) = \sum_j a_j \varepsilon_j \phi_j(y).$$

For $y \in K$, every bump vanishes, so $S(y) = y$. If $y \in Q \setminus K$, one of the covering bumps is positive; all coefficients are positive, so $\theta(y) > 0$. Since $y \neq k_0$, this implies $S(y) \neq y$. Thus $\text{Fix } S = K$ on Q .

Finally set $R = S \circ P_Q$. Nonexpansiveness of P_Q transfers the L_0 -Hölder estimate to all of H . If $R(x) = x$, then $x \in Q$, so $P_Q x = x$ and $S(x) = x$, hence $x \in K$. Conversely every point of K is fixed. This proves (4.3).

If any map fixes K and has Hölder constant B , then for distinct $k, l \in K$ one has $\|k - l\| \leq B\|k - l\|^\gamma$, so $B \geq \|k - l\|^{1-\gamma}$. Taking the supremum over K gives $B \geq D_K^{1-\gamma}$ and proves optimality. $\square$

Theorem 4.5 (Exact finite-dimensional fiber classification). *Fix $n \geq 1$ and the parameters $\lambda, L > 0$, $0 < \gamma < 1$. A set $K \subset \mathbb{R}^n$ is the complete inverse fiber $F^{-1}(0)$ of some graph-maximal RL relation if and only if*

$$K \neq \emptyset, \quad K \text{ is compact}, \quad \text{diam } K \leq \mathcal{R}. \quad (4.5)$$

A set is the complete forward fiber $F(0)$ of such a relation if and only if it is nonempty, compact, and has diameter at most $\mathcal{R}/\lambda$. Translations give the corresponding statements for any specified base input or output.

Proof. Necessity follows from Theorem 4.1 and Lemma 2.4. For sufficiency in the inverse case, take the map R from Theorem 4.4. Its Hölder constant is $(\text{diam } K)^{1-\gamma} \leq L$. Use $C = R$ with full domain in (2.2). Lemma 2.2 gives an RL relation maximal at the specified parameters, even if its least possible Hölder constant is smaller than L . Its output is zero exactly when $p = R(p)$, and at such a parameter its input equals p . Therefore $F^{-1}(0) = \text{Fix } R = K$.

For a forward fiber K with diameter at most $\mathcal{R}/\lambda$, apply Theorem 4.4 to λK to obtain S with $\text{Fix } S = \lambda K$ and constant at most L . Use $C = -S$ in (2.2). The input is zero exactly when $p = S(p)$, and then the output is p/λ . Hence $F(0) = K$. Translating either original coordinate does not change graph differences in the RL inequality, proving the last assertion. $\square$

4.3 A single-valued Lipschitz realization

Corollary 4.6. *Under (4.5), for every $0 < \eta < 1$ the inverse fiber can be realized by a single-valued globally Lipschitz map $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that is graph-maximal at the same parameters and satisfies $F^{-1}(0) = K$. Writing $Q = \overline{\text{conv}}K$ and $D_K = \text{diam } K$, it may be chosen so that*

$$\text{Lip } F \leq \frac{1 + \eta}{\lambda(1 - \eta)}, \quad \sup_u \left\| F(u) - \frac{u - P_Q u}{\lambda} \right\| \leq \frac{\eta D_K}{\lambda}. \quad (4.6)$$

The least Hölder constant of its Cayley map remains $D_K^{1-\gamma}$ (zero for a singleton).

Proof. For $D_K = 0$, with $K = \{k_0\}$ take $F(u) = (u - k_0)/\lambda$. All assertions follow directly. For $D_K > 0$, in (4.4) impose in addition $\varepsilon_p \leq \eta/M_p$. This preserves all previous conclusions. With $E = S - \text{Id}_Q$ the same positive combination gives $\text{Lip } E \leq \eta$ and $\sup_{y \in Q} \|E(y)\| \leq \eta D_K$. The projection case $Q = K$ is included by taking $E = 0$. Thus the realized map can be written $R = P_Q + E \circ P_Q$.

Define $G = (\text{Id} + R)/2$. Projection onto a closed convex set is monotone and nonexpansive. Consequently

$$2\langle G(a) - G(b), a - b \rangle = \|a - b\|^2 + \langle P_Q a - P_Q b, a - b \rangle + \langle E(P_Q a) - E(P_Q b), a - b \rangle \geq (1 - \eta)\|a - b\|^2.$$

Also $\text{Lip } G \leq (2 + \eta)/2$. Put $c = (1 - \eta)/2$ and $M = (2 + \eta)/2$. For any target u and any $0 < t < 2c/M^2$, the map $a \mapsto a - t(G(a) - u)$ is a contraction, since the square of its Lipschitz modulus is at most $1 - 2tc + t^2 M^2 < 1$. The fixed-point theorem therefore shows that G is bijective. Strong monotonicity and Cauchy–Schwarz give $\text{Lip } G^{-1} \leq 2/(1 - \eta)$.

The relation generated by R is thus the single-valued map

$$F(u) = \frac{a - R(a)}{2\lambda} = \frac{a - u}{\lambda}, \quad a = G^{-1}(u).$$

Since $\text{Id} - P_Q$ is nonexpansive,

$$\|(a - R(a)) - (b - R(b))\| \leq (1 + \eta)\|a - b\|,$$

which together with the inverse bound gives the first part of (4.6). Maximality and the exact zero set were proved in Theorem 4.5. Since R fixes K , the optimality argument of Theorem 4.4 gives its least Hölder constant.

It remains to verify the uniform approximation. Set $G_0 = (\text{Id} + P_Q)/2$. Its strong-monotonicity modulus is at least $1/2$, and its inverse is $G_0^{-1}(u) = 2u - P_Q u$. Indeed the projection characterization says $\langle u - P_Q u, y - P_Q u \rangle \leq 0$ for every $y \in Q$; multiplying by 2 shows $P_Q(2u - P_Q u) = P_Q u$. For $a = G^{-1}(u)$ and $a_0 = G_0^{-1}(u)$,

$$\|a - a_0\| \leq 2\|G_0(a) - G_0(a_0)\| = \|E(P_Q a)\| \leq \eta D_K.$$

Since $(u - P_Q u)/\lambda = (a_0 - u)/\lambda$, subtraction gives the second inequality of (4.6). $\square$

In particular, nonconvex compact zero sets can occur in a single-valued globally Lipschitz realization arbitrarily close to the monotone projection residual $(\text{Id} - P_Q)/\lambda$. This does not make that realization strongly monotone: a strongly monotone map has at most one zero.

5 Sharp range localization in global and fixed windows

Lemma 5.1 (Scalar localization). *For $t \geq 0$, let $\rho(t)$ be the largest nonnegative solution of*

$$\rho - t = L(\rho + t)^\gamma. \quad (5.1)$$

Then $\rho(0) = \mathcal{R}$, ρ is continuous and strictly increasing, and its range is $[\mathcal{R}, \infty)$. For every graph anchor $(x_0, v_0) \in \text{gph } F$, every graph point satisfies

$$\|x - x_0\| \leq \rho(\lambda\|v - v_0\|), \quad \lambda\|v - v_0\| \leq \rho(\|x - x_0\|). \quad (5.2)$$

Proof. Write $u = \rho + t$. Equation (5.1) is $u - Lu^\gamma = 2t$. On $u \geq \mathcal{R}$, the left side has derivative $1 - \gamma Lu^{\gamma-1} \geq 1 - \gamma > 0$, starts at zero, and tends to infinity. It thus defines a unique continuous branch $u(t) \geq \mathcal{R}$. Its derivative is $u'(t) = 2/(1 - \gamma Lu(t)^{\gamma-1}) > 2$, so $\rho(t) = u(t) - t$ is strictly increasing and unbounded. For $t > 0$ a nonnegative solution of (5.1) must have $\rho > t$ and $u - Lu^\gamma > 0$, hence $u > \mathcal{R}$, so this solution is unique. For $t = 0$ the possible roots are 0 and $\mathcal{R}$, and the stated choice is the larger root.

For $s = \|x - x_0\|$ and $t = \lambda\|v - v_0\|$, the reverse and ordinary triangle inequalities give $|s - t| \leq L(s + t)^\gamma$. If $s \leq t$, then $s \leq \rho(t)$ because $\rho(t) \geq t$. If $s > t$, the inequality says $(s + t) - L(s + t)^\gamma \leq 2t$. When $s + t \leq \mathcal{R}$ the desired bound is automatic from $\rho(t) \geq \mathcal{R}$; when $s + t > \mathcal{R}$, monotonicity of the preceding scalar function gives $s + t \leq u(t)$, hence $s \leq \rho(t)$. Interchanging s, t proves the other bound. $\square$

For $r > \mathcal{R}$ define $h(r) \in (0, r)$ by

$$r - h(r) = L(r + h(r))^\gamma. \quad (5.3)$$

The left side minus the right side is strictly decreasing as a function of h , positive at $h = 0$ and negative at $h = r$. Thus the root is unique. Lemma 5.1 shows $\rho(h(r)) = r$.

Theorem 5.2 (Global coverage and entire-fiber containment). *Let F be graph-maximal on $\mathbb{R}^n$, and fix $(x_0, v_0) \in \text{gph } F$. For $r > \mathcal{R}$,*

$$B(v_0, h(r)/\lambda) \subset F(B(x_0, r)), \quad F^{-1}(v) \subset B(x_0, r) \quad \text{for every } v \in B(v_0, h(r)/\lambda). \quad (5.4)$$

In the second assertion the inverse fiber is nonempty. The symmetric statement is

$$B(x_0, h(\lambda s)) \subset F^{-1}(B(v_0, s)), \quad F(x) \subset B(v_0, s) \quad (x \in B(x_0, h(\lambda s))),$$

whenever $\lambda s > \mathcal{R}$.

Proof. Theorem 4.1 guarantees a nonempty inverse fiber for every v . If $\lambda\|v - v_0\| < h(r)$, then every point in that fiber satisfies

$$\|x - x_0\| \leq \rho(\lambda\|v - v_0\|) < \rho(h(r)) = r.$$

This gives both coverage and full-fiber containment. For the symmetric statement, full domain gives $F(x) \neq \emptyset$ and the second inequality of (5.2) gives $\lambda\|v - v_0\| < \rho(h(\lambda s)) = \lambda s$ for every $v \in F(x)$. $\square$

Theorem 5.3 (Fixed-window coverage). *Let $U, W \subset \mathbb{R}^n$ and let $G \subset U \times W$ be a nonempty relatively maximal RL graph. Suppose $(x_0, v_0) \in G$, $B(x_0, r) \subset U$, $B(v_0, s) \subset W$, and $r > \mathcal{R}$, $s > 0$. Then every*

$$v \in B(v_0, \min\{s, h(r)/\lambda\}) \quad (5.5)$$

has a nonempty inverse fiber in G , and all its inverse points in G belong to $B(x_0, r)$. In particular, the target ball in (5.5) is contained in the range of the restriction of G to $B(x_0, r) \times W$.

Proof. Apply Lemma 2.3 to the partial Cayley map of G and let $\hat{F}$ be the resulting global graph-maximal completion. The graph $\text{gph } \hat{F} \cap (U \times W)$ contains G and is compatible at the same parameters. Relative maximality therefore gives

$$G = \text{gph } \hat{F} \cap (U \times W).$$

For a target in (5.5), Theorem 5.2 gives a nonempty complete inverse fiber of $\hat{F}$, entirely in $B(x_0, r) \subset U$. The target belongs to W , so the equality of graph restrictions puts every one of these points in G . Conversely, every point of G is in the completion, and is localized by the same theorem. This proves all assertions. $\square$

Proposition 5.4 (Sharp radius and threshold). *The radius $h(r)/\lambda$ in (5.4) cannot be increased uniformly, and the strict threshold $r > \mathcal{R}$ cannot be replaced by $r \leq \mathcal{R}$ while retaining a positive centered output ball uniformly over the class.*

Proof. In dimension one, take $C(p) = L(p_+)^{\gamma}$, where $p_+ = \max\{p, 0\}$. The inequality $|a^{\gamma} - b^{\gamma}| \leq |a - b|^{\gamma}$ for $a, b \geq 0$, together with nonexpansiveness of $p \mapsto p_+$, makes C L -Hölder. Its full graph produces a maximal RL relation anchored at $(0, 0)$, with

$$x(p) = \frac{p + L(p_+)^{\gamma}}{2}, \quad v(p) = \frac{p - L(p_+)^{\gamma}}{2\lambda}.$$

A positive output requires $p > \mathcal{R}$, since for $p \leq 0$ the output is nonpositive and for $0 < p \leq \mathcal{R}$ one has $p \leq Lp^{\gamma}$. On $p > \mathcal{R}$, v is strictly increasing because $v'(p) = (1 - \gamma L p^{\gamma-1})/(2\lambda) > 0$, and it ranges over all positive outputs. The corresponding $x(p)$ is strictly greater than $\mathcal{R}$. Hence no positive output has an inverse in $(-r, r)$ when $r \leq \mathcal{R}$.

For $r > \mathcal{R}$, the parameter $p = r + h(r)$ satisfies $Lp^{\gamma} = r - h(r)$, so $x(p) = r$ and $v(p) = h(r)/\lambda$. This positive output has no other inverse, and its only inverse is outside the open input ball. Any strictly larger centered output radius would include this target and fail. The global example is also a permitted fixed-window example by taking the windows to be the full space. $\square$

6 Finite-data shadows and coverage certificates

The full-graph theorem is an existence theorem: an arbitrary extension does not by itself supply a finite numerical evaluation oracle. We now give a separate construction from a fixed finite data set in $\mathbb{R}^n$. The resulting map is defined globally, but its accuracy for an unknown original relation is certified only at observations and where coverage is available.

6.1 A globally consistent quadratic-program construction

Fix $m \geq 1$ graph samples (x_i, v_i) and set $p_i = x_i + \lambda v_i$, $c_i = x_i - \lambda v_i$. Repeated parameter points in consistent data have the same c_i and may be merged. Fix $0 < \sigma < 1$, and put

$$M = M_{\sigma}, \quad a^2 = M/2, \quad P = [p_1 \ \cdots \ p_m], \quad V = [c_1 \ \cdots \ c_m]. \quad (6.1)$$

The construction requires only the sample compatibility inequalities

$$\|c_i - c_j\|^2 \leq \sigma^2 \|p_i - p_j\|^2 + M. \quad (6.2)$$

Certified Hölder data satisfy these by Lemma 3.1. Define

$$Q = \sigma^2 P^T P + V^T V + a^2 I_m, \quad (6.3)$$

$$d_i(q) = \sigma^2 \|p_i\|^2 - \|c_i\|^2 - 4\sigma^2 \langle p_i, q \rangle, \quad (6.4)$$

$$\Delta_m = \{\theta \in \mathbb{R}^m : \theta_i \geq 0, \sum_i \theta_i = 1\},$$

$$\theta(q) = \underset{\theta \in \Delta_m}{\operatorname{argmin}} \Phi_q(\theta), \quad \Phi_q(\theta) = \theta^T Q \theta + d(q)^T \theta, \quad N_m(q) = V\theta(q). \quad (6.5)$$

Theorem 6.1 (Finite-data contraction and sampled cross estimate). *The minimizer in (6.5) exists and is unique. The formula defines one global σ -Lipschitz map $N_m : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Under (6.2), it satisfies*

$$\|c_i - N_m(q)\|^2 \leq \sigma^2 \|p_i - q\|^2 + a^2 \quad (i = 1, \dots, m, q \in \mathbb{R}^n). \quad (6.6)$$

The graph parametrization

$$x = \frac{q + N_m(q)}{2}, \quad A_m(x) = \frac{q - N_m(q)}{2\lambda} \quad (6.7)$$

defines a strongly monotone bi-Lipschitz homeomorphism with the bounds (3.4)–(3.5). It obeys the cross inequality (3.6) for every sampled graph point and all z , and therefore the paired sampled bounds

$$\|v_i - A_m(x_i)\| \leq \frac{a}{\lambda\sqrt{1-\sigma^2}}, \quad \|x_i - A_m^{-1}(v_i)\| \leq \frac{a}{\sqrt{1-\sigma^2}}. \quad (6.8)$$

Proof. The simplex is compact and nonempty, and $Q \succeq a^2 I_m$ with $a > 0$. Thus the continuous objective is strictly convex and has a unique minimizer.

Let $r = a/\sigma$, introduce lifted points $z_i = (p_i, r c_i)$ in $E = \mathbb{R}^n \oplus \mathbb{R}^m$, and let Z have these columns. Then $Q = \sigma^2 Z^T Z + V^T V$. For every $z \in E$, consider the same minimization with coefficients

$$d_i^E(z) = \sigma^2 \|z_i\|^2 - \|c_i\|^2 - 4\sigma^2 \langle z_i, z \rangle,$$

and denote its unique minimizer by θ_z . At $z = (q, 0)$ the coefficients differ from (6.4) by the same constant a^2 for every i . This changes the simplex objective by a constant, so $\theta_{(q,0)} = \theta(q)$.

For $z, w \in E$, put $\delta = \theta_z - \theta_w$. The variational inequalities at the two minimizers, tested against one another, give

$$\delta^T Q \delta \leq 2\sigma^2 \langle Z\delta, z - w \rangle.$$

Indeed subtraction of the gradients uses $d^E(z) - d^E(w) = -4\sigma^2 Z^T(z - w)$ and division by 2 gives the displayed inequality. Hence

$$\begin{aligned} \|V\delta\|^2 &\leq \sigma^2 (2\langle Z\delta, z - w \rangle - \|Z\delta\|^2) \\ &= \sigma^2 (\|z - w\|^2 - \|Z\delta - (z - w)\|^2) \leq \sigma^2 \|z - w\|^2. \end{aligned}$$

Thus $\tilde{N}(z) = V\theta_z$ is σ -Lipschitz, and its restriction $N_m(q) = \tilde{N}(q, 0)$ has the same modulus. This Lipschitz argument did not require sample compatibility.

Compatibility ensures interpolation at the lifted data. At $z = z_j$, the candidate $\theta = e_j$ satisfies, for each i ,

$$[2Qe_j + d^E(z_j)]_i - [2Qe_j + d^E(z_j)]_j = \sigma^2 \|z_i - z_j\|^2 - \|c_i - c_j\|^2 \geq 0.$$

For $i \neq j$, the last inequality is (6.2), since $\sigma^2 \|z_i - z_j\|^2 = \sigma^2 \|p_i - p_j\|^2 + 2a^2$. For $i = j$ it is equality. These inequalities give the simplex first-order optimality condition at e_j , so uniqueness implies $\theta_{z_j} = e_j$ and $\tilde{N}(z_j) = c_j$. Comparing z_j with $(q, 0)$ yields (6.6).

The contraction-to-operator proof of Theorem 3.3, applied to N_m , proves the assertions about A_m . The same difference-of-squares calculation applied to (6.6) gives (3.6) on sampled graph points, and setting the original input or output equal gives (6.8). $\square$

The quadratic program is also obtained by specializing the explicit Kirszbraun formula of [1] to the finitely many lifted data. The preceding proof is self-contained for this formula and does not require evaluating a convex envelope. The sample set is fixed: adding observations may change the resulting global map. Independently choosing a feasible ball-intersection value at each query, without cross-query compatibility, is not the construction in (6.5).

6.2 Coverage-dependent paired certificates

Theorem 6.2. *Suppose the samples belong to an RL graph with partial Cayley map C . For $\delta \geq 0$ put $b = L\delta^\gamma$ and*

$$K_\delta = \frac{b + \sigma^2\delta + \sqrt{\sigma^2(b + \delta)^2 + (1 - \sigma^2)a^2}}{1 - \sigma^2}. \quad (6.9)$$

Every original graph point (x, v) whose parameter $p = x + \lambda v$ satisfies $\text{dist}(p, \{p_1, \dots, p_m\}) \leq \delta$ obeys

$$\lambda\|v - A_m(x)\| \leq K_\delta, \quad \|x - A_m^{-1}(v)\| \leq K_\delta. \quad (6.10)$$

At $\delta = 0$, these are the bounds in (6.8); if also $\sigma = \sqrt{\gamma}$, $K_0 = \mathcal{R}/\sqrt{2}$.

Proof. Choose a closest sample p_i , which exists because the set is finite. For every $q \in \mathbb{R}^n$, Hölder continuity and (6.6) give

$$\|C(p) - N_m(q)\| \leq b + \sqrt{\sigma^2(\|p - q\| + \delta)^2 + a^2}. \quad (6.11)$$

First take $q = x + \lambda A_m(x)$, so the compared points have the same original input. Then $\|p - q\| = \|C(p) - N_m(q)\| = \lambda\|v - A_m(x)\|$. For the inverse comparison choose $q = A_m^{-1}(v) + \lambda v$; both norms instead equal $\|x - A_m^{-1}(v)\|$. Thus in either case the nonnegative quantity t to be bounded satisfies

$$t \leq b + \sqrt{\sigma^2(t + \delta)^2 + a^2}.$$

If $t < b$, then $t \leq K_\delta$ immediately, since $K_\delta \geq b$. If $t \geq b$, squaring the nonnegative sides gives

$$(1 - \sigma^2)t^2 - 2(b + \sigma^2\delta)t + b^2 - \sigma^2\delta^2 - a^2 \leq 0.$$

The upper root of this upward quadratic is (6.9), because

$$(b + \sigma^2\delta)^2 - (1 - \sigma^2)(b^2 - \sigma^2\delta^2 - a^2) = \sigma^2(b + \delta)^2 + (1 - \sigma^2)a^2.$$

Therefore $t \leq K_\delta$. Substituting $\delta = 0$ proves the final assertions. $\square$

Corollary 6.3 (Entire fibers on a covered region). *Let $\Omega \subset \text{dom } C$ be a region covered within distance δ by the sample parameters. If every point of a nonempty fiber $F(x)$ has parameter in Ω , then $d_H(F(x), \{A_m(x)\}) \leq K_\delta/\lambda$. If every point of a nonempty inverse fiber $F^{-1}(v)$ has parameter in Ω , then $d_H(F^{-1}(v), \{A_m^{-1}(v)\}) \leq K_\delta$.*

Proof. Apply Theorem 6.2 to every graph point in the indicated fiber and use the singleton formula for Hausdorff distance. $\square$

The condition is coverage of $p = x + \lambda v$, not coverage of x alone. For an explicit sufficient parameter radius, suppose $C(0)$ is known and the complete Cayley map is defined. If either $\|x\| \leq B$ or $\lambda\|v\| \leq B$, then

$$\|p\| \leq T_B := \max\{4B + 2\|C(0)\|, (2L)^{1/(1-\gamma)}\}. \quad (6.12)$$

Indeed, from either $p = 2x - C(p)$ or $p = 2\lambda v + C(p)$, $t = \|p\| \leq 2B + \|C(0)\| + Lt^\gamma$. If $t > T_B$, then $2B + \|C(0)\| < t/2$ and $Lt^\gamma < t/2$, a contradiction. Thus a parameter δ -net of $\overline{B}(0, T_B)$ certifies all fibers in either specified original-coordinate working region. Access to such parameter samples is an additional sampling assumption, not a consequence of having one arbitrarily chosen value at each sampled original input.

6.3 Finite-precision evaluation

Proposition 6.4 (Computable evaluation certificates). *For a feasible $\hat{\theta} \in \Delta_m$, let $g_q = 2Q\hat{\theta} + d(q)$ and define its Frank–Wolfe gap*

$$G = \langle g_q, \hat{\theta} \rangle - \min_i \langle g_q, e_i \rangle.$$

Then $G \geq 0$ and

$$\|V\hat{\theta} - N_m(q)\| \leq \sqrt{G}. \quad (6.13)$$

More generally, if $\hat{N}(q)$ approximates $N_m(q)$ with certified error at most e , then any candidate parameter q gives

$$\left\| A_m(x) - \frac{q - x}{\lambda} \right\| \leq \frac{\|q - 2x + \hat{N}(q)\| + e}{\lambda(1 - \sigma)}, \quad (6.14)$$

$$\left\| A_m^{-1}(v) - (q - \lambda v) \right\| \leq \frac{\|q - 2\lambda v - \hat{N}(q)\| + e}{1 - \sigma}. \quad (6.15)$$

Proof. The minimum component of g_q is its minimum over the simplex, so the gap is nonnegative. Convexity gives $\Phi_q(\hat{\theta}) - \Phi_q(\theta(q)) \leq G$. The exact quadratic expansion, together with the first-order inequality at $\theta(q)$, gives

$$\Phi_q(\hat{\theta}) - \Phi_q(\theta(q)) \geq (\hat{\theta} - \theta(q))^T Q(\hat{\theta} - \theta(q)) \geq \|V\hat{\theta} - N_m(q)\|^2,$$

proving (6.13).

If T is a contraction of modulus σ with fixed point q_* , then $\|q - q_*\| \leq \|q - T(q)\| + \sigma\|q - q_*\|$, hence $\|q - q_*\| \leq \|q - T(q)\|/(1 - \sigma)$. Use $T(q) = 2x - N_m(q)$ or $T(q) = 2\lambda v + N_m(q)$, bound the exact residual by its approximate residual plus e , and use $A_m(x) = (q_* - x)/\lambda$ or $A_m^{-1}(v) = q_* - \lambda v$. This gives (6.14)–(6.15). $\square$

Exact evaluations allow ordinary geometric fixed-point iteration. The certificates above also allow any positive target accuracy using sufficiently accurate quadratic-program solutions. Numerical errors must be added to the geometric bounds in Theorem 6.2. Floating-point gaps need validated tolerances or outward rounding if they are used for computer-assisted rigorous numerical certification. No dimension-independent sample or running-time complexity is asserted; covering a region may require many samples in high dimension.

Corollary 6.5 (Bias, observation noise, and evaluation error). *Let v be a true output and $\tilde{v}$ an observation with $\|\tilde{v} - v\| \leq \eta$. Suppose all points of $F^{-1}(v)$ satisfy the coverage condition of Corollary 6.3. If an evaluated proxy $\hat{x}$ satisfies $\|\hat{x} - A_m^{-1}(\tilde{v})\| \leq e$, then every $x \in F^{-1}(v)$ satisfies*

$$\|x - \hat{x}\| \leq K_\delta + \frac{\lambda(1 + \sigma)}{1 - \sigma}\eta + e. \quad (6.16)$$

Proof. Insert $A_m^{-1}(v)$ and $A_m^{-1}(\tilde{v})$ between x and $\hat{x}$. Bound the three resulting distances by the coverage certificate, the inverse Lipschitz constant in (3.4), and e , respectively. $\square$

The first term is model/coverage bias, the second is noise amplification, and the third is evaluation error. The first does not generally vanish when observation and computation errors vanish.

6.4 Why a finite total transcript cannot certify the whole space

Proposition 6.6 (Finite-information obstruction). *Let $n \geq 1$. Consider deterministic procedures which, given point evaluations of an unknown globally L -Hölder map $C : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of exponent γ , make finitely many possibly adaptive queries and output a single-valued map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with no further access to the oracle. No such procedure can guarantee a finite uniform forward-fiber error for the corresponding relation on all of $\mathbb{R}^n$ for every such C . In particular it cannot guarantee the sharp global radius from finite total information alone.*

Proof. Run a proposed procedure on $C_0 = 0$ and let S contain the finitely many queried points together with 0. Fix a unit vector e and define $C_1(p) = L \text{dist}(p, S)^\gamma e$. Distance to S is 1-Lipschitz, and $|s^\gamma - t^\gamma| \leq |s - t|^\gamma$ for nonnegative s, t . Therefore C_1 is L -Hölder. It vanishes at every queried point. By induction over the adaptive queries, the deterministic procedure receives the same transcript on C_1 as on C_0 and outputs the same map A . Both full-domain maps define graph-maximal RL relations.

The relation for C_0 is $F_0(x) = \{x/\lambda\}$. If the output has no finite uniform error for F_0 , the procedure already fails. Otherwise there is $B < \infty$ with $\|A(x) - x/\lambda\| \leq B$ for every x . For a graph point of the relation associated with C_1 , put $c = C_1(p)$, $x = (p + c)/2$, and $v = (p - c)/(2\lambda)$. Then

$$\|v - A(x)\| \geq \|v - x/\lambda\| - \|A(x) - x/\lambda\| \geq \|c\|/\lambda - B.$$

Since S is finite, $\text{dist}(p, S)$ is unbounded on $\mathbb{R}^n$. The last expression is unbounded, so A has no finite uniform forward error for this second relation. This contradicts the proposed guarantee. $\square$

This statement concerns finite total information, deterministic procedures, an unrestricted Hölder class, and an unbounded domain. It does not cover probabilistic guarantees, continued oracle access, known analytic maps, or approximation on a specified compact region.

7 A deadband inverse problem: what the certificate changes

Let U be an orthogonal 2×2 matrix, let $a > 0$, and put

$$K = U[-a, a]^2, \quad F(x) = x - P_K x, \quad \lambda = 1. \quad (7.1)$$

This is a vector deadband: inputs inside K all produce output zero. It is already a monotone single-valued map, and

$$F^{-1}(0) = K, \quad \text{dist}(x, K) = \|F(x)\|. \quad (7.2)$$

The example is not intended to demonstrate that monotone-operator theory fails to apply. It distinguishes distance to a fixed solution set from stable point reconstruction as the target changes.

Proposition 7.1 (Exact graph coordinates and unavoidable inverse jumps). *The full Cayley map of (7.1) is $C = P_K$. For $\gamma = 1/2$, its least Hölder constant is $L = \sqrt{D}$, where $D = \text{diam } K = 2a\sqrt{2}$, and hence $\mathcal{R} = D$. Let $n_j = Ue_j$ and choose $0 < 4\varepsilon < a$. For the two outputs $w_\pm = \pm\varepsilon n_1$,*

$$F^{-1}(w_\pm) = \{\pm(a + \varepsilon)n_1 + tn_2 : |t| \leq a\}. \quad (7.3)$$

Every pair of exact reconstructions $x_{\pm} \in F^{-1}(w_{\pm})$ satisfies

$$\frac{\|x_+ - x_-\|}{\|w_+ - w_-\|} \geq \frac{a}{\varepsilon} + 1. \quad (7.4)$$

Proof. If $q = P_K x$, then the projection characterization says $\langle x - q, y - q \rangle \leq 0$ for every $y \in K$. Thus $p = 2x - q$ has $P_K p = q$, and $x - F(x) = q = P_K(x + F(x))$. Conversely, for any p set $q = P_K p$ and $x = (p + q)/2$. The same projection characterization, scaled by $1/2$, gives $P_K x = q$ and $F(x) = (p - q)/2$. This proves the full parametrization, hence graph maximality. Nonexpansiveness and the image diameter give

$$\|P_K p - P_K q\| \leq \min\{\|p - q\|, D\} \leq \sqrt{D} \|p - q\|^{1/2}.$$

Two diameter endpoints of K , fixed by P_K , force $L \geq \sqrt{D}$, so the constant is least possible.

In rotated coordinates, each component of F is $f(s) = s - \max\{-a, \min\{s, a\}\}$. For an output $\varepsilon > 0$ its unique scalar inverse is $a + \varepsilon$; for $-\varepsilon$ it is $-a - \varepsilon$; for zero its inverse is $[-a, a]$. This yields (7.3). The difference in the n_1 coordinate of any two chosen inverse points is $2a + 2\varepsilon$. Their Euclidean distance is at least that amount, while $\|w_+ - w_-\| = 2\varepsilon$, proving (7.4). $\square$

For a transparent analytic proxy, take the contraction $N = \frac{1}{2}P_K$ and define A by (3.9). Its Lipschitz and inverse Lipschitz constants are at most 3, and its strong-monotonicity modulus is at least $1/3$. This is a model-specific contraction; its choice $\sigma = 1/2$ is not the radius-minimizing parameter $\sqrt{\gamma} = 1/\sqrt{2}$ of the general theorem.

Proposition 7.2 (Stable analytic proxy with full-fiber control). *For the above A , in the rotated coordinates each component of A^{-1} is*

$$\alpha(w) = \begin{cases} 3w, & |w| \leq a/4, \\ w + \frac{a}{2} \operatorname{sign}(w), & |w| \geq a/4. \end{cases} \quad (7.5)$$

The formulas agree at the endpoints. Every complete inverse fiber obeys

$$\sup_{x \in F^{-1}(v)} \|x - A^{-1}(v)\| \leq a\sqrt{2} \quad (v \in \mathbb{R}^2). \quad (7.6)$$

For $w_{\pm}$ from Proposition 7.1,

$$A^{-1}(w_{\pm}) = \pm 3\varepsilon n_1, \quad (7.7)$$

$$\frac{\|A^{-1}(w_+) - A^{-1}(w_-)\|}{\|w_+ - w_-\|} = 3, \quad (7.8)$$

$$\sup_{x \in F^{-1}(w_{\pm})} \|x - A^{-1}(w_{\pm})\| = \sqrt{(a - 2\varepsilon)^2 + a^2}. \quad (7.9)$$

Proof. Writing $\pi(p) = \max\{-a, \min\{p, a\}\}$, scalar inverse evaluation requires $p = 2w + \frac{1}{2}\pi(p)$ and returns $p - w$. If $|w| \leq a/4$, then $p = 4w \in [-a, a]$ solves this equation and returns $3w$. If $w \geq a/4$, then $p = 2w + a/2 \geq a$ solves it and returns $w + a/2$; the negative case is symmetric. The contraction ensures these are the unique solutions, proving (7.5).

For $w > 0$, the scalar true inverse is $a + w$. If $0 < w \leq a/4$, its difference from the proxy is $a - 2w \leq a$; if $w \geq a/4$, its difference is $a/2$. For $w < 0$ the absolute differences are the same. At $w = 0$, the true inverse interval is $[-a, a]$ and the proxy is zero, so the worst difference is a . Applying this bound to each rotated coordinate and using orthogonality gives (7.6). Since $4\varepsilon < a$, the first branch of (7.5) applies to both targets, proving (7.7)–(7.8). Substitution into (7.3) gives the displacement $\pm(a - 2\varepsilon)n_1 + tn_2$; maximizing its norm over $|t| \leq a$ proves (7.9). $\square$

For $a = 1/8$ and $\varepsilon = 10^{-4}$, the output displacement is 0.0002. Every exact inverse pair moves by at least 0.2502, a gain of at least 1251, whereas the analytic proxy moves by 0.0006, a gain of 3. The proxy generally is not an exact inverse point. These numbers compare two different requirements—exact reconstruction and certified approximate reconstruction—and make their tradeoff explicit.

The known analytic model permits other simple choices. In particular, the identity map already gives the global forward and inverse radius $a\sqrt{2}$, because $x - v = P_K x \in K$ and $\sup_{k \in K} \|k\| = a\sqrt{2}$. Thus this example does not establish that the shadow construction dominates bespoke regularization. Its purpose is to exhibit what a common stable proxy and a full-fiber certificate mean, and what they do not mean. For finite graph data, Section 6 supplies a model-class construction and adds the explicit coverage and evaluation terms that an exact analytic calculation does not require.

8 Discussion and scope

The simultaneous shadow theorem and the realization theorem express two complementary conclusions. The entire relation admits one quantitatively controlled, strongly monotone, invertible approximation. Nevertheless, an individual finite-dimensional fiber may be any nonempty compact set of the permitted diameter. The approximation does not turn original fibers into convex sets, select an exact branch, or eliminate nonidentifiability. Its radius is a uniform worst-case guarantee, not a claim that every relation requires the full radius.

The global assumption is restrictive. To see its large-scale meaning, fix a graph anchor and write $d = (x - x_0) + \lambda(v - v_0)$ and $e = (x - x_0) - \lambda(v - v_0)$. Along any graph sequence unbounded in $H \times H$, $\|d\|$ must be unbounded, since $\|e\| \leq L\|d\|^\gamma$ bounds both original coordinate differences whenever d is bounded. Along a sequence with $\|d\| \rightarrow \infty$, one has $\|e\|/\|d\| \rightarrow 0$. Since $x - x_0 = (d + e)/2$, this implies, uniformly for large graph points,

$$v - v_0 = \lambda^{-1}(x - x_0) + O(\|x - x_0\|^\gamma).$$

In particular, the class prescribes a common asymptotic linear slope; it is not a container for every locally regular nonmonotone model.

The sharp constant established here is uniform over dimensions. Determining optimal constants in each fixed dimension, or under additional geometric assumptions on fibers, remains a distinct question. Likewise, a specified anchor changes the approximation problem and its optimal factor. Additional interpolation constraints or stronger computational requirements may impose different tradeoffs.

The finite-data theorem distinguishes a mathematical existence result from a computable surrogate. It supplies one coherent global map from fixed observations and certifies unobserved graph points only where the specified coverage assumption holds. Proposition 6.6 shows why this locality cannot be discarded in the stated oracle model. Efficient adaptive coverage, structured sampling, and dimension-dependent guarantees are natural computational questions, but are not resolved by the existence of a finite quadratic program.

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